

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.063 / 0.0156194
Time	2026-06-28 23:00 UTC
Location	29.7681, -95.2206
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 884	45.96 to 96.77; mean 73.8835	59.12 at 2026-06-28 22:55Z; dt 5 min
asos	temperature	degC	9 / 884	23.3333 to 35; mean 29.582	32 at 2026-06-28 22:55Z; dt 5 min
asos	wind_direction	degree	9 / 1207	0 to 230; mean 162.137	160 at 2026-06-28 22:55Z; dt 5 min
asos	wind_speed	m/s	9 / 1213	0 to 10.8033; mean 4.4794	6.68777 at 2026-06-28 22:55Z; dt 5 min
noaa_gfs	gh_500	m	9 / 108	5909.96 to 5937.36; mean 5924.8	5911.72 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	pbl_height	m	9 / 108	81.3022 to 2281.54; mean 1037.09	1407.46 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	9 / 108	35.7471 to 46.8981; mean 40.9778	42.8682 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	9 / 108	1007.67 to 1016.66; mean 1013	1011.25 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	t_850	degC	9 / 108	17.4979 to 21.7219; mean 19.5482	17.9282 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	u_10m	m/s	9 / 108	-4.54246 to 1.86545; mean -0.7548	-2.02287 at 2026-06-29 00:00Z; dt 60 min
noaa_gfs	v_10m	m/s	9 / 108	2.0274 to 6.81848; mean 4.4087	5.78752 at 2026-06-29 00:00Z; dt 60 min
openaq	ozone	ppm	16 / 1091	-0.001 to 0.069; mean 0.0168	0.063 at 2026-06-28 23:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 215	31 to 127; mean 57.3302	57 at 2026-06-28 23:00Z; dt 0 min
openaq	pm25	ug/m3	81 / 4613	0.1 to 41.4; mean 7.9973	15.2 at 2026-06-28 23:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 54; mean 3.6139	0 at 2026-06-28 22:59Z; dt 0.4 min
openweather	humidity	percent	5 / 360	53 to 92; mean 76.2194	65 at 2026-06-28 22:59Z; dt 0.4 min
openweather	precipitation	mm/h	5 / 360	0 to 0; mean 0	0 at 2026-06-28 22:59Z; dt 0.4 min
openweather	pressure	hPa	5 / 360	1013 to 1017; mean 1015.17	1013 at 2026-06-28 22:59Z; dt 0.4 min
openweather	temperature	degC	5 / 360	24.84 to 35.77; mean 30.2219	33.1 at 2026-06-28 22:59Z; dt 0.4 min
openweather	wind_direction	degree	5 / 360	0 to 254; mean 164.506	137 at 2026-06-28 22:59Z; dt 0.4 min
openweather	wind_speed	m/s	5 / 360	0.45 to 6.6; mean 2.678	3.58 at 2026-06-28 22:59Z; dt 0.4 min
purpleair	pm25	ug/m3	68 / 4715	0 to 65.4; mean 8.9685	9.3 at 2026-06-28 23:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_ch4_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_co_column	mol/m^2	9 / 9	0.0243616 to 0.0276194; mean 0.0263	0.0257712 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_co_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_hcho_column	mol/m^2	7 / 7	8.55934e-05 to 0.000174034; mean 0.0001	8.57164e-05 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_no2_column	mol/m^2	5 / 5	2.61612e-05 to 4.44238e-05; mean 0	2.61612e-05 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_o3_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_so2_column	mol/m^2	7 / 7	-2.15981e-05 to 5.5985e-05; mean 0	-9.57613e-06 at 2026-06-28 20:02Z; dt 177.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-28 20:02Z; dt 177.4 min
tceq	co	ppm	3 / 190	0.1 to 1; mean 0.2137	0.1 at 2026-06-28 23:00Z; dt 0 min
tceq	no2	ppb	11 / 686	-1.7 to 16.6; mean 2.4813	3.4 at 2026-06-28 23:00Z; dt 0 min
tceq	so2	ppb	3 / 189	-1.5 to 0.7; mean -0.3053	-0.2 at 2026-06-28 23:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-27 06Z 90.69, 12Z 70.76, 18Z 54.78, 06-28 00Z 77.58, 06Z 88.78, 12Z 69.01, 18Z 58.13, 06-29 00Z 79.14, 06Z 87.6, 12Z 71.39, 18Z 60.53, 06-30 00Z 78.77, 06Z 88.57
asos	temperature	06-27 06Z 25.26, 12Z 30.15, 18Z 32.7, 06-28 00Z 28.31, 06Z 27.03, 12Z 30.66, 18Z 32.64, 06-29 00Z 28.79, 06Z 27.5, 12Z 30.76, 18Z 32.79, 06-30 00Z 28.13, 06Z 25.76
asos	wind_direction	06-27 06Z 113.5, 12Z 184.7, 18Z 182.8, 06-28 00Z 164.9, 06Z 161, 12Z 192.1, 18Z 174.1, 06-29 00Z 153.8, 06Z 157, 12Z 176.7, 18Z 156.6, 06-30 00Z 141.5, 06Z 98.19
asos	wind_speed	06-27 06Z 1.695, 12Z 4.35, 18Z 5.714, 06-28 00Z 4.476, 06Z 3.443, 12Z 4.931, 18Z 6.138, 06-29 00Z 5.358, 06Z 3.428, 12Z 4.294, 18Z 5.929, 06-30 00Z 3.989, 06Z 1.643
noaa_gfs	gh_500	06-27 12Z 5924, 18Z 5934, 06-28 00Z 5930, 06Z 5929, 12Z 5915, 18Z 5922, 06-29 00Z 5912, 06Z 5920, 12Z 5915, 18Z 5932, 06-30 00Z 5927, 06Z 5936
noaa_gfs	pbl_height	06-27 12Z 416.2, 18Z 1813, 06-28 00Z 1175, 06Z 703.5, 12Z 643.1, 18Z 1764, 06-29 00Z 1175, 06Z 800, 12Z 429.7, 18Z 1860, 06-30 00Z 1081, 06Z 583.8
noaa_gfs	precipitable_water	06-27 12Z 38.44, 18Z 43.73, 06-28 00Z 39.04, 06Z 39.44, 12Z 41.89, 18Z 42.29, 06-29 00Z 42.67, 06Z 41.64, 12Z 37.42, 18Z 43, 06-30 00Z 43.53, 06Z 38.63
noaa_gfs	surface_pressure	06-27 12Z 1014, 18Z 1014, 06-28 00Z 1012, 06Z 1013, 12Z 1013, 18Z 1012, 06-29 00Z 1011, 06Z 1012, 12Z 1013, 18Z 1014, 06-30 00Z 1013, 06Z 1014
noaa_gfs	t_850	06-27 12Z 20.64, 18Z 18.04, 06-28 00Z 19, 06Z 20.55, 12Z 19.33, 18Z 18.47, 06-29 00Z 18.24, 06Z 20.44, 12Z 21.45, 18Z 18.83, 06-30 00Z 19.18, 06Z 20.41
noaa_gfs	u_10m	06-27 12Z 0.2276, 18Z 0.5168, 06-28 00Z -0.9569, 06Z -0.7227, 12Z 1.003, 18Z 0.7199, 06-29 00Z -2.315, 06Z -0.9503, 12Z -0.03739, 18Z -1.494, 06-30 00Z -3.37, 06Z -1.679

Source	Metric	Values
noaa_gfs	v_10m	06-27 12Z 2.33, 18Z 5.586, 06-28 00Z 5.27, 06Z 4.607, 12Z 3.823, 18Z 5.953, 06-29 00Z 5.616, 06Z 4.819, 12Z 2.545, 18Z 4.626, 06-30 00Z 4.606, 06Z 3.122
openaq	ozone	06-27 06Z 0.00575, 12Z 0.01486, 18Z 0.02149, 06-28 00Z 0.01622, 06Z 0.01297, 12Z 0.01946, 18Z 0.02165, 06-29 00Z 0.01741, 06Z 0.01489, 12Z 0.01541, 18Z 0.02264, 06-30 00Z 0.01618, 06Z 0.008343
openaq	pm10	06-27 06Z 44.67, 12Z 57.22, 18Z 57.56, 06-28 00Z 40.94, 06Z 39.78, 12Z 49.56, 18Z 59.89, 06-29 00Z 48.83, 06Z 53.17, 12Z 82.56, 18Z 100.3, 06-30 00Z 52.78, 06Z 45.31
openaq	pm25	06-27 06Z 7.065, 12Z 7.552, 18Z 6.181, 06-28 00Z 6.381, 06Z 7.562, 12Z 8.406, 18Z 7.829, 06-29 00Z 7.859, 06Z 9.706, 12Z 10.53, 18Z 9.471, 06-30 00Z 7.747, 06Z 7.118
openweather	cloud_cover	06-27 06Z 13.25, 12Z 8.677, 18Z 7.833, 06-28 00Z 0.7241, 06Z 3.194, 12Z 2.2, 18Z 0.3667, 06-29 00Z 4.367, 06Z 0.1667, 12Z 7.276, 18Z 3.133, 06-30 00Z 2.516, 06Z 1.12
openweather	humidity	06-27 06Z 89.25, 12Z 78.32, 18Z 60.93, 06-28 00Z 75.21, 06Z 87.32, 12Z 76.43, 18Z 63.93, 06-29 00Z 78.2, 06Z 87.43, 12Z 78.66, 18Z 63.53, 06-30 00Z 77.42, 06Z 86.84
openweather	precipitation	06-27 06Z 0, 12Z 0, 18Z 0, 06-28 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-29 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-30 00Z 0, 06Z 0
openweather	pressure	06-27 06Z 1016, 12Z 1017, 18Z 1015, 06-28 00Z 1015, 06Z 1015, 12Z 1015, 18Z 1014, 06-29 00Z 1014, 06Z 1014, 12Z 1016, 18Z 1015, 06-30 00Z 1016, 06Z 1016
openweather	temperature	06-27 06Z 25.49, 12Z 29.68, 18Z 33.87, 06-28 00Z 29.43, 06Z 27.37, 12Z 30.46, 18Z 33.77, 06-29 00Z 29.55, 06Z 27.89, 12Z 30.53, 18Z 34.35, 06-30 00Z 29.38, 06Z 26.51
openweather	wind_direction	06-27 06Z 153.5, 12Z 186.8, 18Z 185.9, 06-28 00Z 158.9, 06Z 158.3, 12Z 189.7, 18Z 165.9, 06-29 00Z 152.1, 06Z 143.5, 12Z 183.3, 18Z 170.5, 06-30 00Z 136.7, 06Z 140.6
openweather	wind_speed	06-27 06Z 2.092, 12Z 2.285, 18Z 3.26, 06-28 00Z 2.596, 06Z 2.337, 12Z 2.971, 18Z 3.374, 06-29 00Z 2.901, 06Z 2.176, 12Z 2.4, 18Z 3.151, 06-30 00Z 2.737, 06Z 1.907
purpleair	pm25	06-27 06Z 8.599, 12Z 8.33, 18Z 5.839, 06-28 00Z 7.04, 06Z 9.464, 12Z 9.636, 18Z 8.545, 06-29 00Z 9.145, 06Z 11.55, 12Z 11.87, 18Z 9.955, 06-30 00Z 8.181, 06Z 8.037
sentinel5p	s5p_aer_ai_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_ch4_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_co_column	06-27 18Z 0.02567, 06-28 18Z 0.02576, 06-29 18Z 0.02751
sentinel5p	s5p_co_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_hcho_column	06-27 18Z 0.0001396, 06-28 18Z 8.565e-05, 06-29 18Z 0.0001376
sentinel5p	s5p_hcho_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_no2_column	06-27 18Z 2.932e-05, 06-28 18Z 2.616e-05, 06-29 18Z 4.141e-05
sentinel5p	s5p_no2_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_o3_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
sentinel5p	s5p_so2_column	06-27 18Z 1.054e-05, 06-28 18Z -7.765e-06, 06-29 18Z -1.321e-05
sentinel5p	s5p_so2_granule_available	06-27 18Z 1, 06-28 18Z 1, 06-29 18Z 1
tceq	co	06-27 06Z 0.3333, 12Z 0.2611, 18Z 0.2611, 06-28 00Z 0.26, 06Z 0.1778, 12Z 0.2222, 18Z 0.1667, 06-29 00Z 0.1778, 06Z 0.1778, 12Z 0.2278, 18Z 0.1944, 06-30 00Z 0.1778, 06Z 0.2167
tceq	no2	06-27 06Z 5.045, 12Z 2.635, 18Z 1.305, 06-28 00Z 1.968, 06Z 1.34, 12Z 0.953, 18Z 0.803, 06-29 00Z 1.652, 06Z 2.705, 12Z 3.973, 18Z 3.095, 06-30 00Z 3.333, 06Z 5.112
tceq	so2	06-27 06Z -0.5333, 12Z -0.4722, 18Z -0.4667, 06-28 00Z -0.3333, 06Z -0.3222, 12Z -0.2667, 18Z -0.25, 06-29 00Z -0.2333, 06Z -0.2944, 12Z -0.2556, 18Z -0.2111, 06-30 00Z -0.1222, 06Z -0.2889

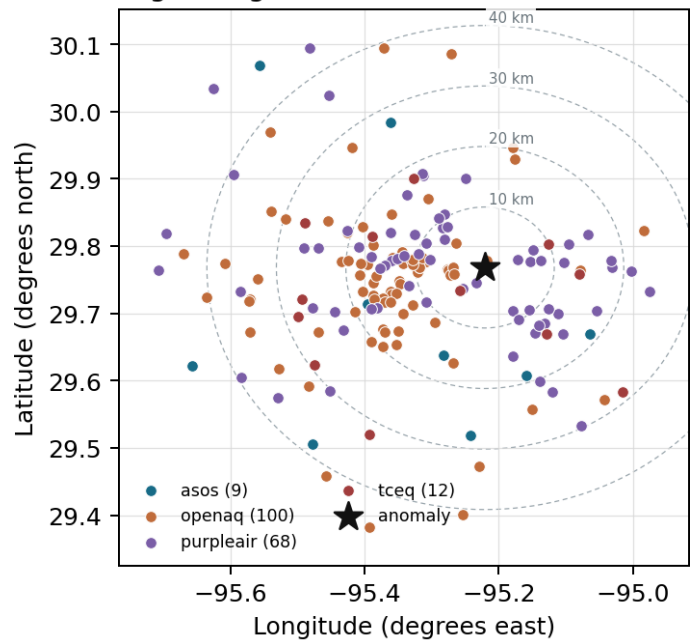
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

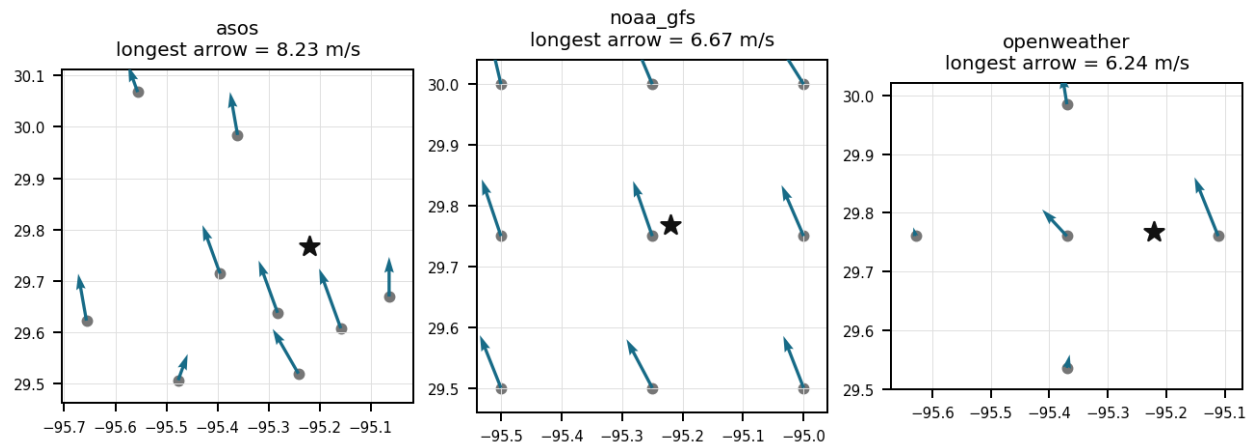
Source	Metric	Values
asos	humidity	HOU (15.701 km) 72.31, MCJ (17.88 km) 75.21, T41 (18.691 km) 73.94, EFD (18.853 km) 74.58, IAH (27.585 km) 72.03, LVJ (27.785 km) 73.36, AXH (38.24 km) 72.17, SGR (45.118 km) 78.54, +1 more
asos	temperature	HOU (15.701 km) 30.09, MCJ (17.88 km) 29.31, T41 (18.691 km) 29.88, EFD (18.853 km) 29.79, IAH (27.585 km) 30.02, LVJ (27.785 km) 29.64, AXH (38.24 km) 29.25, SGR (45.118 km) 29.07, +1 more
asos	wind_direction	HOU (15.701 km) 172, MCJ (17.88 km) 168, T41 (18.691 km) 178.4, EFD (18.853 km) 147.5, IAH (27.585 km) 169.8, LVJ (27.785 km) 155.2, AXH (38.24 km) 126.3, SGR (45.118 km) 170.6, +1 more
asos	wind_speed	HOU (15.701 km) 5.152, MCJ (17.88 km) 5.993, T41 (18.691 km) 5.496, EFD (18.853 km) 5.107, IAH (27.585 km) 4.526, LVJ (27.785 km) 3.651, AXH (38.24 km) 2.382, SGR (45.118 km) 5.066, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.25 (3.479 km) 5925, gfs:29.75,-95.00 (21.39 km) 5925, gfs:30.00,-95.25 (25.941 km) 5926, gfs:29.75,-95.50 (27.046 km) 5924, gfs:29.50,-95.25 (29.946 km) 5924, gfs:30.00,-95.00 (33.425 km) 5927, gfs:29.50,-95.00 (36.651 km) 5924, gfs:30.00,-95.50 (37.29 km) 5925, +1 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (3.479 km) 1209, gfs:29.75,-95.00 (21.39 km) 1018, gfs:30.00,-95.25 (25.941 km) 1259, gfs:29.75,-95.50 (27.046 km) 1164, gfs:29.50,-95.25 (29.946 km) 841.2, gfs:30.00,-95.00 (33.425 km) 769.8, gfs:29.50,-95.00 (36.651 km) 916.2, gfs:30.00,-95.50 (37.29 km) 1324, +1 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (3.479 km) 41, gfs:29.75,-95.00 (21.39 km) 41.48, gfs:30.00,-95.25 (25.941 km) 40.71, gfs:29.75,-95.50 (27.046 km) 40.48, gfs:29.50,-95.25 (29.946 km) 41.22, gfs:30.00,-95.00 (33.425 km) 41.34, gfs:29.50,-95.00 (36.651 km) 41.45, gfs:30.00,-95.50 (37.29 km) 40.32, +1 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (3.479 km) 1014, gfs:29.75,-95.00 (21.39 km) 1014, gfs:30.00,-95.25 (25.941 km) 1012, gfs:29.75,-95.50 (27.046 km) 1012, gfs:29.50,-95.25 (29.946 km) 1014, gfs:30.00,-95.00 (33.425 km) 1013, gfs:29.50,-95.00 (36.651 km) 1015, gfs:30.00,-95.50 (37.29 km) 1010, +1 more
noaa_gfs	t_850	gfs:29.75,-95.25 (3.479 km) 19.38, gfs:29.75,-95.00 (21.39 km) 19.55, gfs:30.00,-95.25 (25.941 km) 19.54, gfs:29.75,-95.50 (27.046 km) 19.5, gfs:29.50,-95.25 (29.946 km) 19.55, gfs:30.00,-95.00 (33.425 km) 19.44, gfs:29.50,-95.00 (36.651 km) 19.74, gfs:30.00,-95.50 (37.29 km) 19.61, +1 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (3.479 km) -0.4399, gfs:29.75,-95.00 (21.39 km) -0.9049, gfs:30.00,-95.25 (25.941 km) -0.7174, gfs:29.75,-95.50 (27.046 km) -0.6816, gfs:29.50,-95.25 (29.946 km) -0.8891, gfs:30.00,-95.00 (33.425 km) -1.132, gfs:29.50,-95.00 (36.651 km) -0.7291, gfs:30.00,-95.50 (37.29 km) -0.2457, +1 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (3.479 km) 4.309, gfs:29.75,-95.00 (21.39 km) 4.505, gfs:30.00,-95.25 (25.941 km) 4.201, gfs:29.75,-95.50 (27.046 km) 4.271, gfs:29.50,-95.25 (29.946 km) 4.317, gfs:30.00,-95.00 (33.425 km) 4.343, gfs:29.50,-95.00 (36.651 km) 5.082, gfs:30.00,-95.50 (37.29 km) 4.459, +1 more
openaq	ozone	2039 (0.0 km) 0.01881, 3378 (5.211 km) 0.01379, 2046 (9.065 km) 0.01741, 3214 (9.934 km) 0.01761, 2800 (13.761 km) 0.01796, 332 (14.072 km) 0.01984, 320 (16.472 km) 0.01254, 1319333 (16.917 km) 0.01899, +8 more
openaq	pm10	13380082 (5.211 km) 56.28, 15852419 (9.388 km) 68.32, 271 (14.072 km) 47.08
openaq	pm25	13787195 (1.131 km) 9.758, 9201803 (4.501 km) 8.932, 9489522 (4.603 km) 7.499, 8967065 (5.194 km) 8.143, 3379 (5.211 km) 15.39, 15539682 (5.23 km) 9.051, 8967123 (5.244 km) 10.41, 8967479 (5.244 km) 9.06, +73 more
openweather	cloud_cover	grid:east (10.604 km) 4.014, grid:center (14.427 km) 2.667, grid:north (28.083 km) 5.319, grid:south (29.582 km) 2.056, grid:west (39.382 km) 4.014
openweather	humidity	grid:east (10.604 km) 79.44, grid:center (14.427 km) 75.94, grid:north (28.083 km) 74.32, grid:south (29.582 km) 76.04, grid:west (39.382 km) 75.35
openweather	precipitation	grid:east (10.604 km) 0, grid:center (14.427 km) 0, grid:north (28.083 km) 0, grid:south (29.582 km) 0, grid:west (39.382 km) 0
openweather	pressure	grid:east (10.604 km) 1015, grid:center (14.427 km) 1015, grid:north (28.083 km) 1015, grid:south (29.582 km) 1015, grid:west (39.382 km) 1015
openweather	temperature	grid:east (10.604 km) 30.13, grid:center (14.427 km) 30.45, grid:north (28.083 km) 30.42, grid:south (29.582 km) 30, grid:west (39.382 km) 30.11
openweather	wind_direction	grid:east (10.604 km) 171, grid:center (14.427 km) 163.6, grid:north (28.083 km) 136.8, grid:south (29.582 km) 184.2, grid:west (39.382 km) 166.9
openweather	wind_speed	grid:east (10.604 km) 4.559, grid:center (14.427 km) 2.913, grid:north (28.083 km) 2.702, grid:south (29.582 km) 1.644, grid:west (39.382 km) 1.572

Source	Metric	Values
purpleair	pm25	166435 (2.815 km) 14.18, 99797 (4.634 km) 8.233, 243843 (4.892 km) 15.22, 166441 (6.532 km) 8.021, 127441 (7.393 km) 11.06, 186327 (7.519 km) 9.375, 222601 (7.938 km) 4.861, 186271 (8.063 km) 10.93, +60 more
tceq	co	482011035 (5.231 km) 0.1476, 482011039 (14.096 km) 0.1672, 482011052 (16.937 km) 0.327
tceq	no2	482011035 (5.231 km) 3.927, 482010026 (9.95 km) 3.708, 482011015 (13.673 km) 2.118, 482011039 (14.096 km) 1.745, 482011052 (16.937 km) 6.024, 482010024 (17.947 km) 1.452, 482011066 (26.769 km) 1.974, 482010047 (26.932 km) 2.853, +3 more
tceq	so2	482011035 (5.231 km) -0.2381, 482011039 (14.096 km) 0.08413, 482010051 (29.284 km) -0.7619

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

OpenAQ recorded a severe surface ozone anomaly of 0.063 ppm in Houston, Texas at 23:00 UTC on June 28, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 35

The increase in PM2.5 levels over the past 24 hours is a likely cause of the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 35

Southeasterly winds transported nitrogen oxide precursors across the local area, sustaining photochemical ozone production.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 35

A fresh primary combustion or sulfur-rich industrial plume is contradicted because the anomaly coincided with low co-pollutant indicators, including carbon monoxide near 0.1 ppm, nitrogen dioxide near 3.4 ppb, sulfur dioxide near -0.2 ppb, and no satellite enhancement of sulfur dioxide or carbon monoxide over the area on the same afternoon.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 35

The openaq ozone context within 50 km from 2026-06-27T11:00:00+00:00 to 2026-06-30T11:00:00+00:00 ranged from -0.001 ppm to 0.069 ppm with a mean of 0.0168 ppm, placing the 0.063 ppm reading near the top of observed values in the window.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 35

The openaq sensor at 29.768, -95.221 recorded ozone of 0.063 ppm at 2026-06-28T23:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 35

Regional photochemical ozone production is strongly supported because eastern Houston was hot, nearly cloud-free, and dry enough for efficient photochemistry at the anomaly time, with about 32 to 33.1 degrees Celsius temperature, about 59 to 65 percent relative humidity, zero precipitation, and 0 percent cloud cover near 23:00 UTC on 2026-06-28.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 35

High surface temperatures exceeding 32 degrees Celsius combined with near-zero cloud cover on the afternoon of June 28, 2026, provided conditions for rapid photochemical ozone production.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 35

Prevailing southeasterly surface winds transported nitrogen dioxide and other chemical precursors across the region toward the anomaly site.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 35

Deep planetary boundary layer growth reaching over 1,700 meters during the afternoon promoted vertical atmospheric mixing of ozone and precursor species.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 35

The temperature increase in the past 24 hours is a contributing factor to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 35

Nearby trace-gas observations do not support a fresh combustion or sulfur-rich plume as the dominant cause because contemporaneous TCEQ measurements near the anomaly were about 0.1 ppm CO, 3.4 ppb NO₂, and -0.2 ppb SO₂, so the most likely cause is secondary photochemical ozone formed upwind and transported into eastern Houston.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 35

Transport of an ozone-rich precursor-aged air mass into eastern Houston is supported because surface winds near the anomaly were southerly to southeasterly at about 137 to 160 degrees with moderate speeds of about 3.6 to 6.7 meters per second, which is consistent with advection from the Gulf and ship-channel corridor rather than stagnation.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 35

The ozone anomaly at 29.768, -95.221 had an expected value of 0.015619354838709677 ppm and a z-score of 7.3617484853196045, which was classified as severe by openaq.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 35

The openaq ozone monitor at eastern Houston recorded 0.063 ppm at 2026-06-28T23:00:00+00:00, which was far above the site's expected 0.0156 ppm and above the context-window mean of 0.0168 ppm, indicating an exceptional local ozone enhancement.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 35

The combination of high PM2.5 and NO2 levels is likely the primary cause of the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 35

Clear skies with near-zero cloud cover and afternoon temperatures exceeding 32°C created optimal conditions for photochemical ozone formation on June 28, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 19 of 35

Hot temperatures, very low cloud cover, no precipitation, and a deep afternoon boundary layer on 2026-06-28 favored strong photochemical ozone production near eastern Houston.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 20 of 35

Deep daytime boundary layer mixing exceeding 1700 meters combined with southeasterly winds near 6 m/s facilitated the dispersion and reaction of ozone precursor emissions.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 21 of 35

High solar irradiance under near-zero cloud cover and ambient temperatures exceeding 32 °C accelerated photochemical ozone formation during the afternoon of June 28, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 22 of 35

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 23 of 35

Meteorology near eastern Houston at the anomaly time favored strong ozone production because air temperatures were about 32–33 C, cloud cover was 0%, precipitation was absent, surface winds were from the south to southeast at roughly 3.6–6.7 m/s, and GFS boundary-layer height near 00Z was about 1407 m.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 35

Southerly to southeasterly winds on 2026-06-28 were consistent with transport of ozone and its precursors from the Houston Ship Channel, petrochemical corridor, and coastal urban-industrial areas toward the anomaly location in eastern Houston.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 35

The ozone anomaly is less consistent with a fresh local combustion plume and more consistent with aged photochemical air or recirculated air because co, no2, and so2 near the event were low while ozone was strongly elevated.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 35

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 35

The recorded ozone concentration reached 0.063 ppm at 2026-06-28T23:00:00+00:00, significantly exceeding the expected baseline value of approximately 0.0156 ppm with a z-score of 7.36.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 35

Deep vertical mixing within a daytime planetary boundary layer reaching over 1,700 meters facilitated the photochemical accumulation of surface ozone prior to the evening boundary layer transition.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 35

The wind direction and speed changes over the past 24 hours may have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 35

NO2 levels are also significantly higher than normal during this period.

Mark exactly one:

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Note:

Claim 33 of 35

A severe surface ozone anomaly occurred at coordinates (29.768, -95.221) in Houston, Texas, on 2026-06-28 at 23:00:00 UTC.

Mark exactly one:

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Note:

Claim 34 of 35

The PM2.5 levels are elevated, with a mean of 8.9685 ug/m3 and a maximum value of 65.4 ug/m3.

Mark exactly one:

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Note:

Claim 35 of 35

The anomaly occurred on June 28th, with the highest PM2.5 levels recorded at 23:00:01+00:00.

Mark exactly one:

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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
